

Planar Bilipschitz Extension from Separated Nets: Literature Status

fa_banach_001, agent_lane_01

2026-07-18

Status

Literature already answered, two-dimensional subproblem. This note records that the planar case of the bilipschitz extension problem for separated nets, flagged through Dymond–Kaluza arXiv:2507.22007, is answered by the separate companion paper *Planar bilipschitz extension from separated nets*, Journal of the London Mathematical Society 113 (2026), e70540.

Source Question

Dymond and Kaluza, *Extending Bilipschitz Mappings between Separated Nets*, arXiv:2507.22007, recall the Alestalo–Trotsenko–Vaisala question: for $d \geq 2$, does every bilipschitz mapping from $\mathbb{Z}^d$ to $\mathbb{R}^d$, and more generally from a separated net of $\mathbb{R}^d$ to $\mathbb{R}^d$, admit a bilipschitz extension to all of $\mathbb{R}^d$? In the source, Theorem 1.1 reduces the arbitrary separated-net problem to the integer-lattice problem in every finite dimension.

Answering Paper

The companion article Dymond–Kaluza, *Planar bilipschitz extension from separated nets*, JAMS 113 (2026), e70540, DOI: 10.1112/jlms.70540, states as Theorem 1.1 that every L -bilipschitz mapping $f : \mathbb{Z}^2 \rightarrow \mathbb{R}^2$ extends to a $p(L)$ -bilipschitz mapping $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ for a polynomial p . Combining this with the reduction theorem from arXiv:2507.22007 gives Corollary 1.2: for every separated net $X \subseteq \mathbb{R}^2$ and every L -bilipschitz $f : X \rightarrow \mathbb{R}^2$, there is a bilipschitz extension $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with a bound depending polynomially on L and on the separation/net constants of X .

The supporting paper explicitly states that this answers Navas’ 2015 Oberwolfach problem and gives the positive two-dimensional solution of the Alestalo–Trotsenko–Vaisala problem.

Scope

This packet does not claim any new proof. It records an already-known literature answer for $d = 2$. The higher-dimensional problem for $d \geq 3$, including the hyperplane Conjectures 1 and 2 in arXiv:2507.22007, remains open from the source’s perspective.

Evidence

The source arXiv PDF is included as `source_paper.pdf`. The supporting JLMS DOI page was checked on 2026-07-18; direct PDF download was blocked by a Cloudflare challenge, saved as `supporting_paper_jlms_2026_download_blocked.html`. Web metadata from Wiley and ISTA Research Explorer both record the 2026 JLMS publication and the explicit planar theorem.

References

- [1] M. Dymond and V. Kaluza, *Extending Bilipschitz Mappings between Separated Nets*, arXiv:2507.22007; Annales Fennici Mathematici 51 (2026), 237–260.
- [2] M. Dymond and V. Kaluza, *Planar bilipschitz extension from separated nets*, Journal of the London Mathematical Society 113 (2026), e70540. DOI: 10.1112/jlms.70540.